UNIT – I

Introduction to PHP

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In this Unit, following points to be discussed

01 Introduction of the Unit

02 Evolution of PHP

03 Basic Syntax

04 Defining Variable and Constant

05 PHP Data Types

06 Operator and Expression

07 Conclusion of Unit

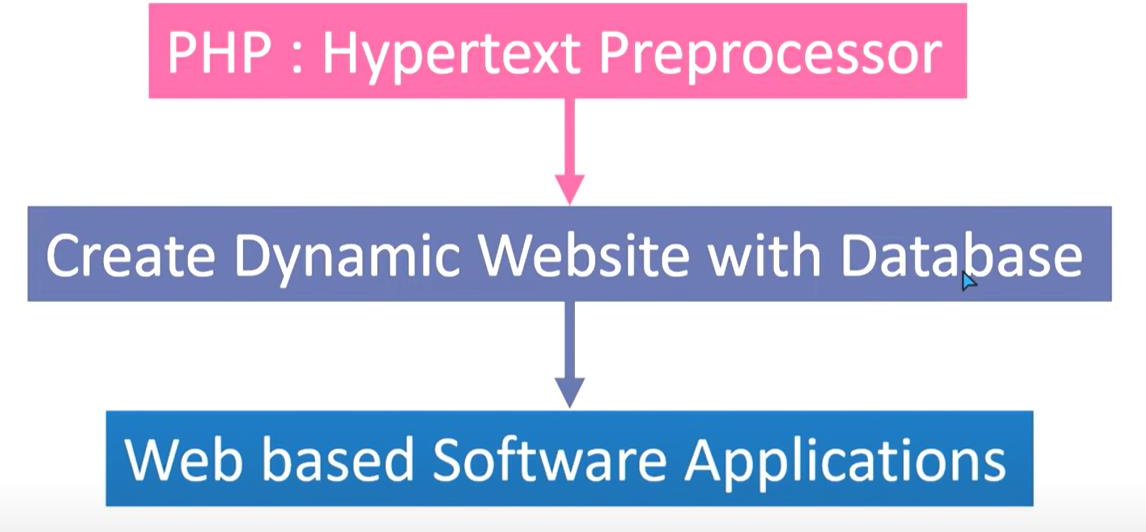
01 Introduction to the Unit

- **PHP** or "**PHP: Hypertext Preprocessor**" is a widely-used **Open Source, Server-Side scripting language** that is especially suited for **Web development** and can be embedded into HTML.

- The main goal of the language is to allow web developers to write dynamically and generate web pages quickly.

- **MySQL** is a RDBMS (Relational database management system). It is used to store, retrieve and manage data related to our applications.
- A **web-server** is required to host a webpage on a remote server and is responsible for processing the request from the client, who wants to access the webpage.
- **Apache** is an Open Source web server .

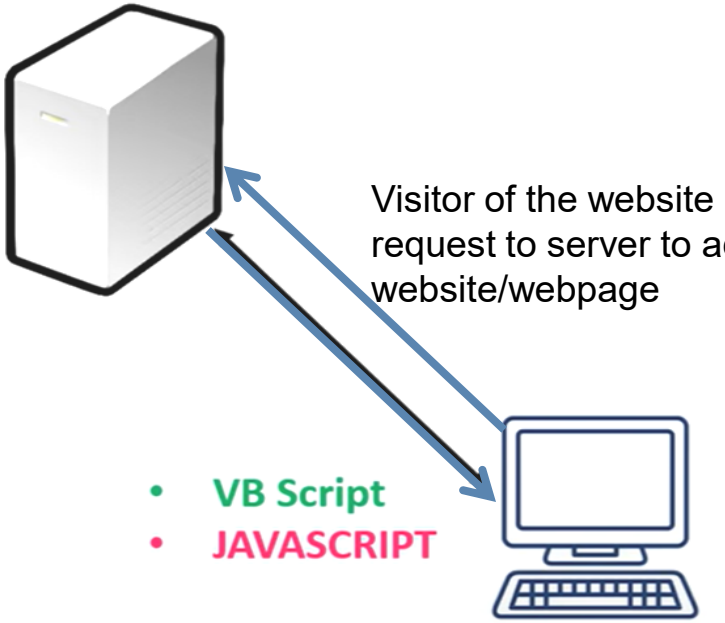
What is a PHP File?



- PHP files can contain Text, HTML, CSS, JavaScript, and PHP code.
- PHP code is executed on the server, and the result is returned to the browser as plain HTML
- PHP files have extension ".php"

Web based Scripting Language

Client Side Script

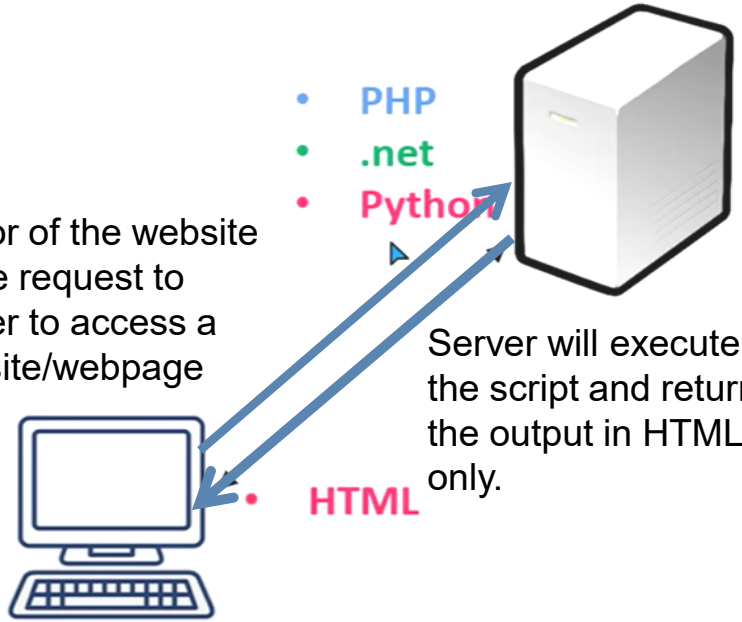


Visitor of the website made request to server to access a website/webpage

- VB Script
- JAVASCRIPT

Server will return the file consisting CSS+Scipt+HTML to the client's browser and then it will run by the client's browser.

Server Side Script



- PHP
- .net
- Python

Visitor of the website made request to server to access a website/webpage

Server will execute the script and return the output in HTML only.

- HTML

Difference between Server-Side Scripting and Client-Side Scripting

1. Client-side scripting :

- Web browsers execute client-side scripting.
- It is used when browsers have all code.
- Source code is used to transfer from webserver to user's computer over the internet and run directly on browsers.
- It is also used for validations and functionality for user events.
- It allows for more interactivity.
- It cannot be basically used to connect to databases on a web server.
- These scripts cannot access the file system that resides in the web browser.
- Pages are altered on basis of the user's choice.
- It can also be used to create "cookies" that store data on the user's computer.

2. Server-side scripting:

- Web servers are used to execute server-side scripting.
- They are basically used to create dynamic pages.
- It can also access the file system residing at the webserver.
- A server-side environment that runs on a scripting language is a web server.
- Scripts can be written in any of a number of server-side scripting languages available.
- It is used to retrieve and generate content for dynamic pages.
- In this load times are generally faster than client-side scripting.
- When you need to store and retrieve information, a database will be used to contain data.

What Can PHP Do?

- PHP can generate dynamic page content
- PHP can create, open, read, write, delete, and close files on the server
- PHP can send and receive ‘cookies’
- PHP can add, delete, modify data in your database
- PHP can be used to control user-access
- PHP can encrypt data

With PHP we are not limited to output HTML. We can output images or PDF files.

Why Use PHP?

- Ease of use and learn
- Cost Efficiency (Open Source). PHP is free. Download it from the official PHP resource: www.php.net
- Object Oriented Programming
- Cross-platform compatibility means PHP can runs on various platforms (Windows, Linux, Unix, Mac OS X, etc.)
- PHP is compatible with almost all servers used today (Apache, IIS, etc.)
- Secure due to multilayer security system.
- Huge Community
- PHP supports a wide range of databases

Popular Websites developed in PHP

- Facebook (i.e. Social Media)
- Yahoo (i.e. Search Engine)
- Wikipedia
- Wordpress (i.e. CMS based Software)
- MailChimp (i.e. Email Marketing Website)
- Flickr (i.e. Image Hosting Website)
- Flipkart (i.e. E-Commerce Website)

Use of PHP in Web Development

- CMS (Content Managment System)
- E-Commerce Website
- Mail Server
- Chatting System
- File Storage System (eg. Google Drive)
- ERP (Enterprise Resource Planning)
- CRM (Customer Relationship Managment)

Prior knowledge before Learning PHP

- HTML
- CSS (Basics)
- JavaScript (Optional)

Software Requirement for PHP

- **HTML Editor**
- **Web Browser**
- Wamp or Xampp **Server**

Popular Editors for PHP

- Notepad
- Notepad ++
- VS Code
- Sublime
- Atom
- PhpStorm (Paid Software)

Assignment

Q1. Diffentiate between Client Side Scripting and Server Side Scripting.

Q2. What are the software requirements for PHP and enlist the popular editors used to write PHP code.

Q3. Enlist various advantages of PHP.

Study Material

S. No	Text Books:	Author	Edition	Publication
1.	PHP: The Complete Reference	Steven Holzner	1 July 2017	TMH
Reference Book				
1.	Learning PHP, MySQL & JavaScript with j Query, CSS & HTML5 – 1 January 2015			
Online Resources				
1.	https://www.w3schools.com/php/			
2.	https://www.tutorialspoint.com/php/index.htm			

Evolution of PHP

02 Evolution of PHP

PHP (Hypertext Preprocessor) has undergone significant evolution since its inception in the mid-1990s.

Milestones in the evolution of PHP:

1. Creation (1994): PHP was created by Rasmus Lerdorf in 1994 as a set of Perl scripts for tracking visits to his online resume. **Originally, it stood for "Personal Home Page," but it later evolved to "PHP: Hypertext Preprocessor.“**

2. PHP/FI (1995): Rasmus Lerdorf released **PHP/FI (Forms Interpreter)** in 1995, which added more functionality and **introduced a form interpreter**.

3. Zend Engine (1999): PHP 4, released in 1999, introduced the Zend Engine, **a scripting engine that greatly improved the performance and capabilities of PHP**. This version marked a significant step forward for the language.

4. PHP 5 (2004): PHP 5, released in 2004, brought several important features, **including the introduction of object-oriented programming (OOP)** with the Zend Engine 2, SimpleXML, SQLite support, and improvements in performance and error handling.

5. PHP 6 (Development Abandoned): PHP 6 development was started with the goal of adding native Unicode support. However, due to technical challenges and delays, the development was eventually abandoned, and many features planned for PHP 6 were backported to PHP 5.3 and later versions.

6. PHP 7 (2015): PHP 7, released in 2015, was a major milestone for the language. It brought significant **performance improvements, the introduction of the spaceship operator (<=>), the scalar type declarations, return type declarations, and more**.

7. PHP 7.1, 7.2, 7.3, 7.4, 8.0 (2016-2020): Subsequent versions of PHP continued to introduce new features, enhancements, and improvements in performance and security. **Notable features include typed properties, arrow functions, JIT (Just-In-Time) compilation in PHP 8.0**, and more.

8. PHP 8.1 (2021): PHP 8.1 brought new features and improvements, building on the foundation laid by PHP 8.0. The exact features will be determined as the release approaches.

Throughout its evolution, PHP has become one of the most popular server-side scripting languages, widely used for web development.

The PHP community, along with the PHP Standards Group (PSR) and the PHP-FIG (PHP Framework Interop Group), actively contributes to the language's growth and standardization.

Company involved in PHP

Many companies use PHP for web development, and PHP is a widely used server-side scripting language.

Organizations that have historically been involved with PHP development:

- **Facebook:** PHP was originally developed by Rasmus Lerdorf in 1994, and Facebook initially used PHP for its platform.
- **WordPress:** The popular content management system (CMS) WordPress is written in PHP. WordPress is used by a large number of websites around the world.
- **Wikipedia:** The backend of Wikipedia is powered by PHP. MediaWiki, the software behind Wikipedia, is written in PHP.
- **Magento:** Magento is an e-commerce platform written in PHP. It is widely used for building online stores.
- **Joomla!:** Another popular CMS, Joomla!, is also written in PHP. It is used for building websites and online applications.

- **Drupal:** Drupal is a powerful **content management system** written in PHP. It's used for building a variety of websites and applications.

- **Zend Technologies:** The company behind the development of the Zend Engine, which is the scripting engine that interprets PHP.

- **Automatic:** The company behind WordPress.com and other products related to WordPress. They heavily contribute to the WordPress ecosystem, which is PHP-based.

- **Etsy:** PHP is used in various parts of Etsy's platform, contributing to the functionality of the popular e-commerce website.

- **Laravel:** Laravel is a **PHP web application framework** that is used by many developers and organizations for building modern, robust web applications.

How to Run PHP

To start using PHP, we can:

- install a web server
- install PHP
- install a database, such as MySQL

The official PHP website (PHP.net) has installation instructions for PHP: http://php.net/manual/en/install.php

Basic Syntax

PHP Syntax

- A PHP script is executed on the server, and the plain HTML result is sent back to the browser.
- A PHP script can be placed anywhere in the document.
- A PHP script starts with <?php and ends with ?>:

```
<?php

    // PHP code goes here

?>
```

- The default file extension for PHP files is ".php".
- A PHP file normally contains HTML tags, and some PHP scripting code.

Example: A simple .php file with both HTML code and PHP code:

```
<html>
<body>

<h1>My first PHP page</h1>

<?php
echo "Hello World!";
?>

</body>
</html>
```

Note: "echo" is a built-in PHP function to output the text "Hello World!"

Remember !!!

- PHP statements ends with a semicolon (;).
- PHP is not case-sensitive.
- All variable names are case-sensitive.

Comments

A comment in PHP code is a line that is not executed as a part of the program. Its only purpose is to be read by someone who is looking at the code.

PHP supports several ways of commenting:

```
// This is a single-line comment

# This is also a single-line comment

/* This is a
multi-line comment */
```

Note: The multi-line comment syntax can also be used to prevent execution of just parts of a code-line:

```
$x = 5 /* + 15 */ + 5;

echo $x;
```

Defining variable and constant

PHP Variables

In PHP, a variable starts with $ sign, folled by the name of the variable,

Example,

```
<?php
    $txt = “Hello World”;
    $x =  5;
    $y = 10.5;
?>
```

Declaring Variables in PHP:

A variable in PHP is a name of memory location that holds data.

OR

A variable is a temporary storage that is used to store data temporarily.

In PHP, a variable is declared using $ sign followed by variable name.

Syntax: **$variablename=value;**

Rules for naming variable:

Valid-
In PHP, variable must starts with letter or underscore (_) only.

$a = “hello”;
$_b = “hello”;

Invalid-
The variable can’t be start with numbers and special symbols.

$4c =“hello”;
$*d = “hello”;

Variable names are case_sensitive, so variable name “A” is different from “a” i.e. $A is not similar to $a

Example -

$A = “Ram”;

$a = “Rohit”;

where both ($A and $a) are different.

PHP is a loosely typed language, it means PHP automatically converts the variable to its correct data type.

Example:

```
$str = “hello”;        // string type
$x = 200;              // integer type
$y = 14.6;             // float type
```

So, There is no need to declare or specify any data type.

Variabledemo.php

```
<html>
  <head>
    <title> Variables Demo </title>
  </head>
     <body>
         <?php
                $str = “Ram”;
                $x = 113;
                $y = 12.23;
                echo “String is : $str
<br/>”;
                echo “Integer is : $x
<br/>”;
                echo “Float is : $y <br/>”;
          ?>
    </body>
</html>
```

Output:

String is : Ram

Integer is : 113

Float is : 12.23

How to write a Variable Name :

Write Way

- $firstname
- $_firstname
- $first_name
- $first-name
- $firstName
- $firstname99

Wrong Way

- $first name
- $99firstname
- $first%name

$age $AGE

Not same

Constant Variables are Global Variables
Don't use $ sign in Constant Variable name.

PHP Constant Variables

```
define(name, value, case-insensitive)
```

```
define(num, 500, true)
echo NUM;
```

Study Material

S. No	Text Books:	Author	Edition	Publication
1.	PHP: The Complete Reference	Steven Holzner	1 July 2017	TMH
Reference Book				
1.	Learning PHP, MySQL & JavaScript with j Query, CSS & HTML5 – 1 January 2015			
Online Resources				
1.	https://www.w3schools.com/php/			
2.	https://www.tutorialspoint.com/php/index.htm			

PHP Data Types

PHP Data types

Data types represent the kind of data a variable can hold.

PHP is a loosely typed language, which means you don't have to declare the data type explicitly — it is determined automatically at runtime.

PHP Supports the following Data Types -

- **String**
- **Integer**
- **Float (Double)**
- **Boolean**
- **Array**
- **Object**
- **NULL**

Getting the Data Type

The var_dump() function returns the data type and the value:

```
<html>
<body>
  <?php
      $x=5;
      var_dump($x);
   ?>
</body>
</html>
```

Output will be : int (5)

1. String

A string is a sequence of characters (text).

Example:

```
$name = "Hello, PHP";
echo $name;
```

output: Hello, PHP

2. Integer

An integer is a whole number (positive or negative) without decimals.

An integer data type is a non-decimal number between -2,147,483,648 and 2,147,483,647.

Rules for integers:

- An integer must have at least one digit
- An integer must not have a decimal point
- An integer can be either positive or negative

Example:

```
$age = 25;
echo $age;
```

Output: 25

3. Float (Double)

A float is a number with a decimal point or in exponential form.

Example:

```
$price = 99.99;
echo $price;
```

Output:99.99

4. Boolean

A boolean can have only two possible values: true or false.

Example:

```
$is_logged_in = true;

echo $is_logged_in;
```

Output: 1

(because true is represented as 1 when printed)

5. Array

An array is a collection of values (like a list).

Example:

```
$fruits = array("Apple", "Banana", "Mango");
echo $fruits[0];
```

Output: Apple (the first item in the array)

6. Object

An object is a data type that stores data and information on how to process it. It is an instance of a class.

Example:

```
class Car {
public $color = "Red";
}
```

// This declares a class named **Car**

// **Public** (to be accessed outside from the class **$color** contains the **property of the class**.

```
$myCar = new Car();
```

i.e.

// **new Car()** makes a new **Object of Class** Car myCar

```
echo $myCar->color;
```

// -> is used to access a property of an object. So, $myCar->color means "get the color of the car".

Output: Red

7. NULL

The NULL data type represents a variable with no value.

Example:

```
$x = NULL;
var_dump($x);
```

Output:NULL

8. Resource

A resource in PHP is a **special variable** that holds a reference to an external resource such as:

- File handling
- Database connections
- Network streams

We can’t manipulate a resource with regular data handling, **PHP functions handle it internally.**

Example:

```
<?php
$handle = fopen("sample.txt", "r");

if ($handle)
{
echo "File opened successfully!<br>";
var_dump($handle);
 fclose($handle);
}
else
 {
    echo "Failed to open file.";
}
?>
```

fopen("sample.txt", "r") opens the file in read mode ("r").

$handle becomes a resource variable that points to the opened file.

var_dump($handle) shows something like:
resource(3) of type (stream)

fclose($handle) properly closes the file.

Data Type	Description	Example Code	Output
String	Text or characters	"Hello"	Hello
Integer	Whole numbers	25	25
Float	Decimal numbers	99.99	99.99
Boolean	True or False	true	1
Array	Collection of values	["A", "B", "C"]	A
Object	Instance of a class	$obj->property	Depends
NULL	No value	NULL	NULL
Resource	Reference to external resources	fopen()	resource(...)

How to change Data Type

- If you assign an integer value to a variable, the type will automatically be an integer.
- If you assign a string to the same variable, the type will change to a string.

Question :

Differentiate between an **Indexed array**, an **Associative array**, and a **Multidimensional array** in PHP.

1. Indexed Array

An indexed array is an array where **each element is automatically assigned a numeric index starting from 0**.

Syntax:

```
$fruits = array("Apple", "Banana", "Mango");
```

Accessing Elements:

```
echo $fruits[0];
```

Output:

Apple

2. Associative Array

An associative array **uses named keys (strings) to store values** instead of numeric indexes.

Syntax:

```
$student = array ("name" => "Rohit", "age" => 21, "course" =>
"B.Tech");
```

Accessing Elements:

```
echo $student["name"];
```

Output:

Rohit

3. Multidimensional Array

- A multidimensional array is an **array containing one or more arrays**.
- It is used to represent tabular or matrix data.

Syntax:

```
$marks = array(
    array("Math", 85),
    array("Science", 90),
    array("English", 88)
);
```

Accessing Elements:

```
echo$marks[1][0];
```

Output:

Science

Operators and Expressions

PhP Operators

PHP divides the operators in the following groups:

- Arithmetic operators
- Assignment operators
- Comparison operators
- Increment/Decrement operators
- Logical operators
- String operators
- Array operators
- Conditional assignment operators

Arithmetic operators

Operator	Name	Example	Result
+	Addition	$x + $y	Sum of $x and $y
-	Subtraction	$x - $y	Difference of $x and $y
*	Multiplication	$x * $y	Product of $x and $y
/	Division	$x / $y	Quotient of $x and $y
%	Modulus	$x % $y	Remainder of $x divided by $y

PHP Assignment Operators

Assignment	Same as	Description
x = y	x = y	Value of y is assigned to x
x += y	x = x + y	Addition
x -= y	x = x - y	Subtraction
x *= y	x = x * y	Multiplication
x /= y	x = x / y	Division
x %= y	x = x % y	Modulus

PHP Comparison Operators

Operator	Name	Example	Result
==	Equal	$x == $y	Returns true if $x is equal to $y
!=	Not equal	$x != $y	Returns true if $x is not equal to $y
<>	Not equal	$x <> $y	Returns true if $x is not equal to $y
>	Greater than	$x > $y	Returns true if $x is greater than $y
<	Less than	$x < $y	Returns true if $x is less than $y
>=	Greater than or equal to	$x >= $y	Returns true if $x is greater than or equal to $y
<=	Less than or equal to	$x <= $y	Returns true if $x is less than or equal to $y
===	Identical	$x === $y	Returns true if $x is equal to $y, and they are of the same type
!==	Not identical	$x !== $y	Returns true if $x is not equal to $y, or they are not of the same type
<=>	Spaceship	$x <=> $y	Returns an integer less than, equal to, or greater than zero, depending on if $x is less than, equal to, or greater than $y. Introduced in PHP 7.

Example (Spaceship Operator):

$x= operand1 <=> operand2

Here,

- **If operand1 is less than operand2, $x will be -1**
- **if both the operands are equal, $x will be 0**
- **if operand1 is greater than operand2, $x will be 1**

PHP Increment / Decrement Operators

Operator	Name	Description
++$x	Pre-increment	Increments $x by one, then returns $x
$x++	Post-increment	Returns $x, then increments $x by one
--$x	Pre-decrement	Decrements $x by one, then returns $x
$x--	Post-decrement	Returns $x, then decrements $x by one

PHP Logical Operators

Operator	Name	Example	Result
and	And	$x and $y	**True**, if both are true
&&	And	$x && $y	**True**, if both are true
or	Or	$x or $y	**True**, if either one is true
||	Or	$x || $y	**True**, if either one is true
!	Not	!$x	**True**, if $x is not true

PHP String Operators

Operator	Name	Example	Result
.	Concatenation	$txt1 . $txt2	Concatenation of $txt1 and $txt2
.=	Concatenation assignment	$txt1 .= $txt2	Appends $txt2 to $txt1

- Concatenation operator ('. ') returns the **concatenation of its right and left arguments.**
- Concatenating assignment operator (' . = ') **appends the argument on the right side**

Example1:

```
$firstName = "Rohit";
$lastName = "Sharma";

$fullName = $firstName . " " .
$lastName;
echo$fullName;
```

Output: Rohit Sharma

Example2:

```
$message = "Hello";

$message .= ", how are you?";

echo $message;
```

Output: Hello, how are you?

PHP Conditional Assignment Operators

1. Ternary Operator:

- The **ternary operator** in PHP is a shorthand for an if-else statement.
- It allows you to assign a value to a variable based on a condition in a single line.

Operator	Name	Example	Result
? :	Ternary	$x = expr1 ? expr2 : expr3	Returns the value of $x.

Value of $x depends on the following conditions-

- If Expr1 is TRUE, the value of $x will be Expr2
- If Expr1 is FALSE, the value of $x will be Exp3

Example -

```
$age = 20;
$status = ($age >= 18) ? "Adult" : "Minor";

echo "Status: $status";
```

Output:
Status: Adult

2. Null Coalescing Operator (??)

- The null coalescing operator (??) in PHP is used to return the first operand if it exists and is not null, otherwise it returns the second operand.
- It is especially useful for providing default values when a variable or array element may be undefined or null.

Syntax:

```
$value = $variable ?? $default;
```

- If $variable is set and not null, $value gets $variable.
- Otherwise, $value gets $default.

??	Null coalescing	$x = expr1 ?? expr2	Returns the value of $x.

Value of $x depends on the following conditions,

- If Expr1 is true and not NULL, the value of $x will be Expr1
- If Expr1 is not true or is NULL, the value of $x will be Exp2

Study Material

S. No	Text Books:	Author	Edition	Publication
1.	PHP: The Complete Reference	Steven Holzner	1 July 2017	TMH
Reference Book				
1.	Learning PHP, MySQL & JavaScript with j Query, CSS & HTML5 – 1 January 2015			
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1.	https://www.w3schools.com/php/			
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